A Risk-Based Two-Factor Student Verification and Event Attendance Management System Using Near Field Communication (NFC), Personal Identification Number (PIN) Authentication, and Quick Response (QR) Credential Validation

Mendoza, Justine G., Milallos, Michael Daniel M., and Sintay, Seanne Stiffel T.
Taguig City University
College of Information and Communication Technology
Taguig City, Philippines
Justine G. Mendoza: [email address]
Michael Daniel M. Milallos: [email address]
Seanne Stiffel T. Sintay: [email address]

Abstract: This study developed and evaluated a risk-based student verification and event attendance management prototype for Taguig City University. An ESP32 and PN532 NFC reader were integrated with a WinUI 3 application, PIN authentication, camera-based QR validation, local MySQL storage, and Firebase Firestore synchronization. Authorized operators selected Fast, Standard, or High-Security modes. A quantitative developmental and descriptive-evaluative design used purposive and convenience sampling. The proposed-system survey involved 70 students, five security personnel, and five faculty/event-organizer/administrator respondents; 70 different students evaluated the manual process. Controlled observation produced 210 successful entries from 70 participants using 20 reusable credential profiles across three modes. Mean internal latency was 11.49, 57.37, and 54.06 milliseconds for Fast, Standard, and High-Security modes. Recorded Standard PIN and High-Security PIN/QR interaction averaged 2.505 and 5.799 seconds. Observed credential-decision accuracy was 100% (222/222) under researcher-labelled controlled conditions. Overall proposed-system survey means were 3.74, 3.63, and 3.81 for the three respondent groups, interpreted as Highly Satisfactory. Student overall ratings exceeded manual-process ratings (3.742 versus 2.284), Welch $t(89.08) = 15.12$, $p < .001$. Manual writing averaged 21.803 seconds, but unequal timing boundaries prevented a matched complete-task speed comparison. Findings support demonstrated prototype functionality and favorable perceived quality within the evaluation setting. Broader security, recovery, and matched performance testing are needed before institutional deployment.

Key Words: *NFC; PIN authentication; QR credential validation; student verification; attendance management*

I. INTRODUCTION

Student verification and event attendance require institutions to connect an access decision with an identifiable, retrievable record. Paper sign-in sheets collect attendance information, but identity checking, enrollment validation, record consolidation, and monitoring

remain separate tasks. These concerns motivated a prototype for Taguig City University that combined configured credential validation with attendance records and monitoring interfaces.

NFC and Internet of Things technologies have been investigated for attendance capture and data analysis. Lim et al. (2023) described an attendance approach integrating NFC and IoT with analytics. Such work provides a basis for electronic record capture, but the presence of a readable card alone does not establish that the person presenting it is its rightful owner. El Gaabouri et al. (2023) reviewed authentication and threat challenges in RFID-based NFC applications, emphasizing the need to consider security beyond identification.

Additional authentication requirements introduce usability and operational demands. Mohammed et al. (2023) discussed multifactor authentication approaches and their security challenges. In the present prototype, NFC supplied the possession-based credential and PIN entry supplied a knowledge-based check where required. QR matching added a check against the stored credential record; it was not treated as an independently established third authentication factor. The software quality evaluation drew on selected characteristics of the ISO/IEC 25010:2011 model (International Organization for Standardization [ISO], 2011).

The study aimed to develop the prototype, document its responses to configured verification conditions, describe recorded performance components, and evaluate perceived quality among its intended users. It also compared overall survey ratings from independent manual-process and proposed-system student groups. The contribution was an operator-configured combination of verification modes and administrative workflows evaluated under controlled conditions, rather than an automatic individual risk-scoring engine or a claim of university-wide readiness.

II. METHODS

Research Design and Participants

A quantitative developmental and descriptive-evaluative design was used. Development integrated the hardware reader, application, credential rules, and record-management functions. The evaluation combined documented functionality cases, controlled system observation, a manual attendance-writing baseline, and role-specific questionnaires.

Purposive and convenience sampling recruited 80 proposed-system evaluators: 70 students from the College of Information and Communication Technology, five security personnel, and five faculty/event-organizer/administrator respondents. A separate group of 70 students evaluated the manual process. The student survey

groups were independent, not matched pairs. The non-probability sample supported findings about the participating groups and did not establish representativeness of the entire university.

Prototype and Verification Workflow

The prototype used an ESP32 microcontroller and PN532 NFC reader connected to a computer through USB serial communication. A C# WinUI 3 application handled credential lookup, PIN entry, camera-based QR scanning, access decisions, transaction records, and monitoring. Local MySQL storage and Firebase Firestore synchronization supported record management.

Fast mode required NFC validation. Standard mode required NFC and PIN, while High-Security mode additionally required the scanned QR credential to match the stored record. Authorized operators selected the mode. The workflow checked configured student status and entry/exit state; event attendance also required registration for the selected event. Successful event verification recorded attendance, whereas student verification updated entry/exit records. Denials produced the configured alert and transaction record.

Sixteen functionality cases were documented in the thesis. They included valid mode workflows, incorrect PIN and lockout handling, QR mismatch, unknown credentials, inactive and expelled status, restricted-event registration, entry/exit rules, and selected offline, synchronization, and application-closure conditions. These cases were evaluated separately from the numerical observation records.

Controlled System Observation

System observation was conducted on August 12 and 13, 2026, with batches of 40 and 30 participants. The same 70 participants tested Fast, Standard, and High-Security modes in that order. Twenty NFC cards with preconfigured PIN and QR credentials were reused. For each mode, the first batch completed two rounds of 20 entries, and the second completed rounds of 20 and 10 entries. Research-team exit transactions reset the profiles between rounds, and explanations preceded mode changes.

Exports contained transaction timestamps, outcomes, internal latency, database-query timings, and recorded PIN/QR durations. All 210 granted participant entries were retained for timing analysis, including slow observations. Reused profile names were not treated as duplicate participants. Team reset exits were excluded from entry timing. Database time overlapped internal latency and was not added to it. Fast-mode PIN/QR duration was not applicable because these stages were bypassed.

Supplementary recorded-entry spacing used the within-round span from the first to the last successful entry. The four rounds

contributed 19 + 19 + 19 + 9 = 66 intervals per mode. Mean spacing was the summed spans divided by 66; the recorded-entry rate was 60 times 66 divided by the summed spans in seconds. Within-round pauses and invalid attempts remained in those spans. No independent complete-round stopwatch times were available.

Credential and State Outcome Classification

Expected outcomes were assigned from the researcher's observation account. Valid participant grants and deliberately invalid credential denials counted as correct decisions when they agreed with the expected outcome. The accuracy denominator excluded team reset exits and pre-authentication entry/exit state violations. Denial proportion used all exported records, including resets and challenge conditions. Newly triggered PIN lockouts were divided by captured individual PIN submissions, including retries, triggering attempts, and correct PIN stages followed by QR denial. Fast mode and reset exits were excluded from PIN submissions.

Manual Baseline and Survey Instruments

The manual simulation produced 70 observations of participants writing their name, course and section, student ID, and signature. Timing began at the first written character and ended when all required information was complete. Personnel did not check an ID or enrollment status during the timed procedure. This measured active writing rather than queue waiting, complete identity verification, or later encoding.

Four-point questionnaires assessed perceived quality. The student and security instruments covered Functional Suitability, Performance Efficiency, Usability, Reliability, and Security. The administrative instrument also included Compatibility, Maintainability, and Portability. Means of 3.26-4.00 were interpreted as Highly Satisfactory, 2.51-3.25 as Satisfactory, 1.76-2.50 as Unsatisfactory, and 1.00-1.75 as Highly Unsatisfactory. Role-specific instruments were analyzed separately rather than pooled into one system-wide score.

Statistical Analysis

Means and sample standard deviations described applicable observations. Individual survey category scores were averages of each respondent's answers in that category; overall scores were the equally weighted average of the five shared categories. Student composite SDs were calculated from respondent-level scores using n - 1, not by averaging item SDs. Composite SDs were not reported for the two five-person groups because only item-frequency totals were supplied.

A two-tailed Welch independent-samples t-test at alpha = .05 compared the overall student survey scores. The null hypothesis stated that the population mean overall ratings were equal; the alternative stated that they differed. Welch's test did

not require equal population variances. Its statistic was calculated as follows:

$$t = \frac{\bar{x}_S - \bar{x}_M}{\sqrt{\frac{{s_S}^2}{n_S} + \frac{{s_M}^2}{n_M}}}$$

The subscripts S and M denote proposed-system and manual groups; x-bar denotes the group mean, s the sample SD, and n the respondent count. Welch-Satterthwaite degrees of freedom and a 95% confidence interval were calculated from unrounded respondent-level scores. Category comparisons and mode timings were descriptive. No processing-time t-test was performed because manual writing and automated component timings had different boundaries.

III. RESULTS

Prototype Functionality

All 16 documented functionality cases were marked PASS. The export audit contained 438 records: 210 successful participant entries, 210 granted research-team reset exits, and 18 denied records. Each mode contributed 70 successful entries and 70 granted resets. Fast, Standard, and High-Security modes contained 2, 12, and 4 denied records, respectively.

Recorded Performance Components

Table 1 summarizes internal processing and interaction components. Standard's internal latency sample SD was 17.46 ms, compared with 2.20 ms for Fast and 6.34 ms for High-Security. Standard's mean PIN duration was 2.505 seconds. High-Security's mean PIN and QR durations were 2.162 and 3.637 seconds, totaling 5.799 seconds. Its combined median was 4.348 seconds and maximum was 22.604 seconds; all successful trials were retained.

Table 1 Successful entry timing components

Mode	Internal latency (ms)	Interaction (s)
Fast	11.49	N/A
Standard	57.37	2.505
High-Security	54.06	5.799

Note. n = 70 per mode. Interaction denotes PIN or combined PIN/QR stages, not the complete task. Fast bypassed these stages.

The summed within-round spans were 180.200, 531.167, and 800.508 seconds for Fast, Standard, and High-Security. With 66 intervals per mode, mean recorded-entry spacing was 2.730, 8.048, and 12.129 seconds. Corresponding supplementary rates were 21.98, 7.46, and 4.95 intervals per minute. These were timestamp-spacing measures, not independently measured complete-task throughput.

Credential Decisions and Denials

The applicable credential sample comprised 210 expected-valid grants and 12 expected-invalid credential denials. All 222 decisions agreed with the

researcher-confirmed expectations (Table 2). The 18 denied records comprised eight PIN failures, five consecutive-entry violations, two newly triggered lockouts, two QR mismatches, and one irregular exit. The five consecutive-entry violations and irregular exit were state tests outside the credential-accuracy denominator.

Both lockouts were deliberate demonstrations reaching the third consecutive incorrect PIN submission. The researcher confirmed that individual PIN submissions were captured. The 152 submissions consisted of 140 PIN stages in successful entries, eight incorrect PIN records, two triggering lockout submissions, and two correct PIN stages followed by QR denial. The challenge records did not represent unexpected ordinary-user failures.

Table 2 Recorded security measures

Measure	Count	Result
Correct credential decisions	222/222	100%
Denied exported records	18/438	4.11%
Newly triggered PIN lockouts	2/152	1.32%

Note. Denominators differ. Denials include deliberate challenges and resets in the record base; lockouts were deliberately triggered.

Manual Attendance Baseline

The 70 active writing durations had a mean of 21.803 seconds, sample SD of 4.436 seconds, median of 21.135 seconds, and range of 13.21-37.63 seconds. Dividing 60 by the mean yielded a service-time-equivalent rate of 2.75 writing completions per minute. This derived value did not include queue waiting, participant transitions, or subsequent encoding and was not a continuously observed throughput rate.

Respondent Evaluations

Overall proposed-system means were 3.74 for students, 3.63 for security personnel, and 3.81 for faculty/event-organizer/administrator respondents, all Highly Satisfactory. The manual student overall mean was 2.28, interpreted as Unsatisfactory. Table 3 gives the respondent-level student category and overall descriptive statistics.

Table 3 Student survey means and sample SDs

Category	Manual M (SD)	System M (SD)
Functional Suitability	2.483 (0.899)	3.763 (0.318)
Performance Efficiency	1.932 (0.919)	3.743 (0.366)
Usability	2.637 (0.854)	3.740 (0.329)
Reliability	2.183 (0.852)	3.711 (0.348)
Security	2.183 (0.883)	3.751 (0.341)
Overall	2.284 (0.753)	3.742 (0.290)

Note. Independent groups, n = 70 each. SDs are respondent-level composite sample SDs. Overall scores use equal category weights.

Security personnel rated Functional Suitability at 3.84,

Performance Efficiency at 3.45, Usability at 3.60, Reliability at 3.48, and Security at 3.80. Administrative respondents rated the same categories at 3.80, 3.68, 3.88, 3.76, and 3.84, respectively, and rated Compatibility at 3.75, Maintainability at 3.88, and Portability at 3.92. All category means in these two proposed-system groups fell within the Highly Satisfactory range.

Independent Student Survey Comparison

The proposed-system and manual groups each contained 70 respondents. Their overall means were 3.742 and 2.284, with sample SDs of 0.290 and 0.753. Using the unrounded scores, the standard error was 0.096411 and the mean difference was 1.458143, giving t = 1.458143 / 0.096411 = 15.124. Table 4 reports the Welch result. The two-tailed p-value was below .001, and the null hypothesis was rejected at .05.

Table 4 Welch test of overall survey scores

Statistic	Result
Mean difference (system - manual)	1.458
t	15.124
Welch df	89.076
Two-tailed p	< .001
95% CI for the difference	[1.267, 1.650]
Decision at alpha = .05	Reject H0

IV. DISCUSSION

Perceived Quality and Practical Use

The independent student groups reported substantially different evaluations, with higher overall ratings for the proposed prototype. The largest descriptive category difference was 1.811 points in Performance Efficiency. Functional Suitability differed by 1.280, Usability by 1.103, Reliability by 1.529, and Security by 1.569 points. These results concerned participants' reported perceptions across corresponding categories. They did not show that every individual would improve by the same amount after changing methods.

The prototype combined credential decisions, timestamped records, and monitoring within the same workflow. This configuration offers a practical basis for organizing access and attendance information, consistent with the record-capture direction discussed by Lim et al. (2023). Nevertheless, this study did not quantify a reduction in transcription errors, proxy attendance, or administrative labor. Favorable ISO-based ratings represent user evaluation rather than formal ISO certification (ISO, 2011).

Security personnel gave their lowest category mean to Performance Efficiency, at 3.45, and administrative respondents also rated that category lowest, at 3.68. These ratings identify monitoring and administrative

workflows for further investigation. They do not by themselves establish whether a particular screen, hardware component, or database operation caused the lower ratings.

Interpretation of Timing Measurements

The internal measurements described processing on a millisecond scale, while the recorded PIN and QR stages occurred on a seconds scale. Added credential interaction therefore needs to be assessed separately from software latency. Standard's mean internal latency exceeded High-Security's, so internal timings did not increase strictly with the number of required credential stages. Standard also had greater internal variability in this observation.

The lower High-Security PIN mean cannot establish that the mode improved PIN performance, because participants always completed Standard before High-Security. Practice, changing conditions, and the demonstration schedule could have influenced the observations. The reused cards and research-team resets also differed from ordinary campus operation.

The manual baseline measured writing, whereas automated exports measured internal and credential-stage components. Comparing 21.803 seconds directly with 2.505 or 5.799 seconds would mix different task boundaries. Similarly, the manual service-time equivalent and automated recorded-entry interval rates were different constructs. The study therefore established neither a matched percentage time reduction nor a complete-task throughput advantage.

Security Scope and Study Limitations

The recorded credential accuracy demonstrated expected grants and correct rejections in the known controlled sample. It did not establish that an NFC UID, PIN, or printed QR code could never be copied or shared. El Gaabouri et al. (2023) identified broader RFID/NFC authentication threats, while Mohammed et al. (2023) emphasized challenges that remain in multifactor systems. The present results support configured rule handling, not comprehensive resistance to those threat classes.

The application label for consecutive-entry denial described an entry-state rule. Without physical passage sensing or a controlled barrier, such a rule does not establish prevention of someone following another person through an entrance. Similarly, a matching QR credential is an additional integrity check against the stored record, not independent evidence of the presenter's physical identity.

Generalization was limited by non-probability sampling, the five-person operational groups, fixed mode order, and 20 reusable credential profiles. Expected outcomes were researcher-labelled rather than independently blinded. Deliberate demonstrations affected

denial and lockout proportions, and exports did not independently audit unrecorded cancellations, missed scans, or complete elapsed verification tasks. The manual and system survey groups differed, and process-specific item wording limited equivalence of measurement. Individual responses were unavailable for composite SDs in the two five-person groups.

Selected offline and synchronization cases passed, but these observations were not a prolonged recovery, load, or network-failure study. The results justify further pilot testing with defined acceptance criteria before institution-wide deployment.

V. CONCLUSION

The study produced a prototype combining NFC capture, PIN authentication, QR credential validation, configured student and event rules, transaction recording, and monitoring. All 16 documented functionality cases passed, and 222 applicable recorded credential decisions agreed with researcher-confirmed expectations. The proposed-system student, security, and administrative groups gave Highly Satisfactory overall evaluations.

The Welch test found higher overall survey ratings in the proposed-system student group than in the separate manual-process group, $t(89.08) = 15.12$, $p < .001$. This supported a difference in reported evaluations within the study setting. Recorded timings provided useful component-level performance evidence, but their boundaries did not establish a matched complete-task speed or throughput advantage. The controlled tests did not establish universal security effectiveness or readiness for immediate campus-wide use.

VI. RECOMMENDATION

1. Measure manual and automated workflows using the same explicit start and finish boundaries. Record waiting, identity checks, participant transitions, failures, and complete-round elapsed time separately. Use these matched observations for an appropriate processing-time comparison.

2. Extend security testing to shared or copied credentials, unknown and inactive profiles, repeated attempts, credential revocation, and recovery from lockout. Use independently specified expected outcomes and assess physical entry controls separately from software entry-state rules.

3. Conduct repeated load, hardware-disconnection, power-interruption, and synchronization tests. Verify record reconciliation, prevention of duplicate records, recovery behavior, and backup restoration under defined failure conditions.

4. Investigate the monitoring workflow and QR interaction experience through direct observation and broader user feedback. Recruit participants from additional colleges and increase

security and administrative sample sizes. Retain individual response records and document instrument comparability before extending inferential claims.

5. Begin operational use with a supervised pilot, staff training, explicit access permissions, privacy and retention procedures, and documented alternatives during outages or exceptional access cases. Evaluate campus-wide readiness against predefined performance and security criteria.

VII. REFERENCES

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